

Memristor Modelling for Common Source Amplifier Using 180 nm Technology

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Abstract

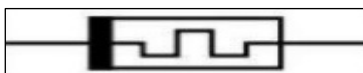
The objective of this research paper is on memristor modeling for common source amplifier circuit using cadence virtuoso tool at gpdk 180 nm technology. A memristor is the fourth element count after resistor, inductor and capacitor in fundamental electrical circuits assumption is proposed by Leon Chua, its characteristics is based on already occurred principles of voltages (V), current (I), flux (ϕ) and charge (q). Ohm's law states relation voltage to current is directly proportional by ($V=IR$), charge is directly proportional to voltage ($q=CV$), and relation between current to flux ($\phi=LI$) and, as above equations represents that connection between flux (ϕ) and the charge (q).

Keywords: GPDK (generic process design kit), TEAM (ThrEshold adaptive memristor), memristors, SMCS (single memristor common source amplifier), DMCS (double memristor common source amplifier)

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INTRODUCTION

The memristor connects the charge and flux, $d\phi=Mq$ an equation-defined relationship between flux and charge, quantity of flowing charge decide its resistance; therefore, it acts like a linear resistor with memory and sometimes having nonlinear characteristics also symbol representation of memristor as [1, 2].



There are various prototypes of electronic schematics with memristor having been demonstrated to describe the electrical characteristic of memristor devices such as the linear ion drift model, the nonlinear ion drift model, Simmons tunnel barrier model, and the ThrEshold Adaptive Memristor (TEAM) model. The present memristor prototype schematic is designed in Cadence Virtuoso with considering of linear drift model of memristor. The schematic test bench of memristor is shown in Figures 1 and 2.

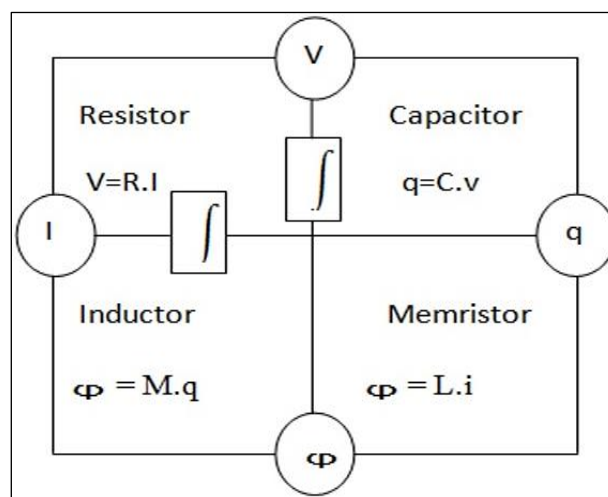


Fig. 1: Relationships between Voltage v, Charge q, Flux ϕ , and Current i [1].

For this modelling, the width of the titanium dioxide TiO_2 film (D) is assumed as 3 nm and assumed width of doped region (w) is as 1 nm. And other parameters such as resistance of doped region, $R_{\text{ON}}=100 \Omega$, and undoped region, $R_{\text{OFF}}=20 \text{ K}\Omega$ verilog code as [3, 4].

Parameters Definitions and Default Values

Parameter real $R_{\text{off}}=200000$;
 Parameter real $R_{\text{on}}=100$;
 Parameter real $D=3 \text{ n}$;
 Parameter real $uv=1\text{e-}15$;
 Parameter real $w_multiplied=1\text{e}8$;

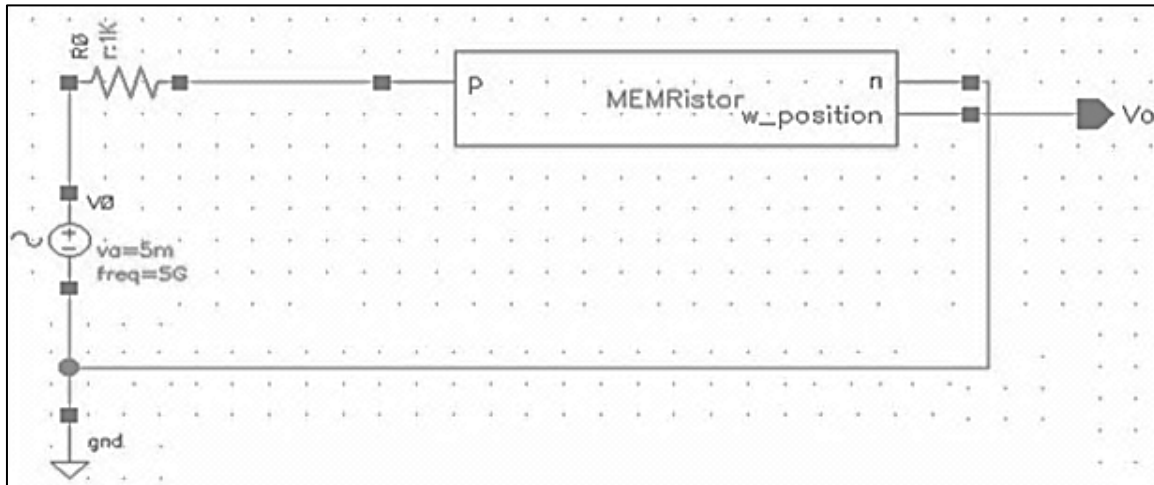


Fig. 2: Schematic Presentation of Memristor with Having Feature of TiO_2 Film $D=3 \text{ nm}$ and as $w=1 \text{ nm}$, $R_{\text{ON}}=100 \Omega$ and $R_{\text{OFF}}=20 \text{ K}\Omega$ [2].

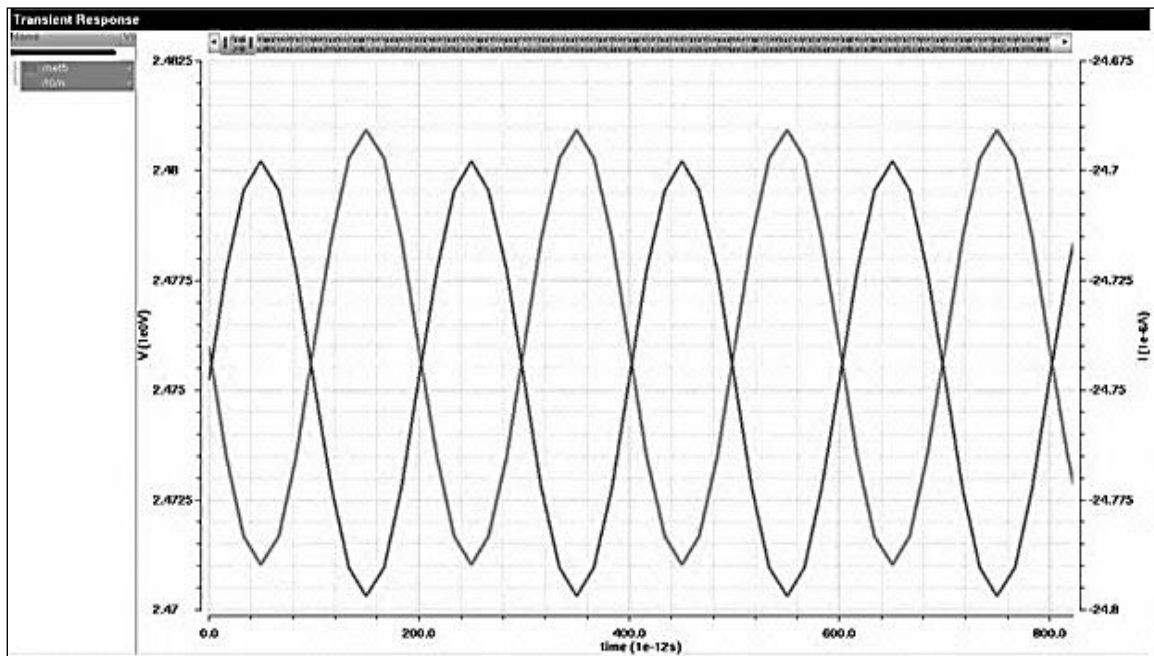


Fig. 3: Voltage, Current vs. Time Characteristics [3].

On pursuing transient analysis at 0 to 40 nsec, we get the characteristics between voltage, current and the time as shown in Figure 3.

As per result analysis of Voltages versus Time characteristics, an average value of voltage is 2.475 V, maximum value of voltage is up to 2.480 V and the minimum value of voltage

2.470V and an average value of current is $-24.47\text{e-}6 \text{ A}$, maximum value of current is up to $-24.69\text{e-}6 \text{ A}$ and the minimum value of current is $-24.79\text{e-}6 \text{ A}$ [4].

Characteristics of voltage and the current with respect to time are shown individually in Figures 4 and 5, respectively.

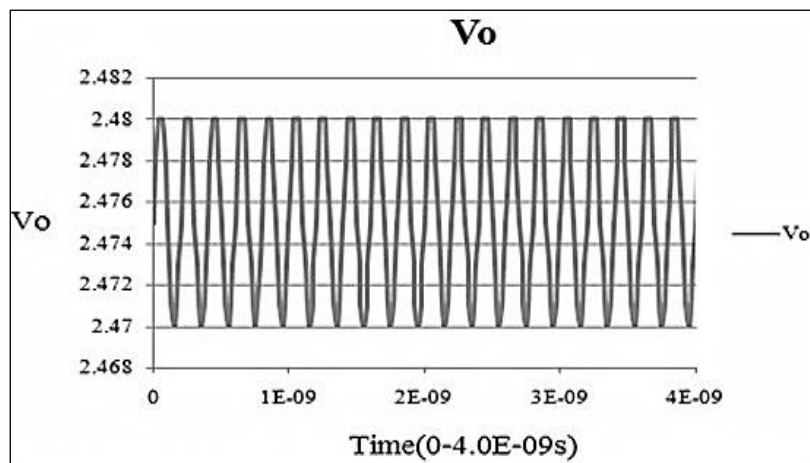


Fig. 4: Voltage vs. Time Characteristics where Analysis Considered up to 40 nsec [4, 5].

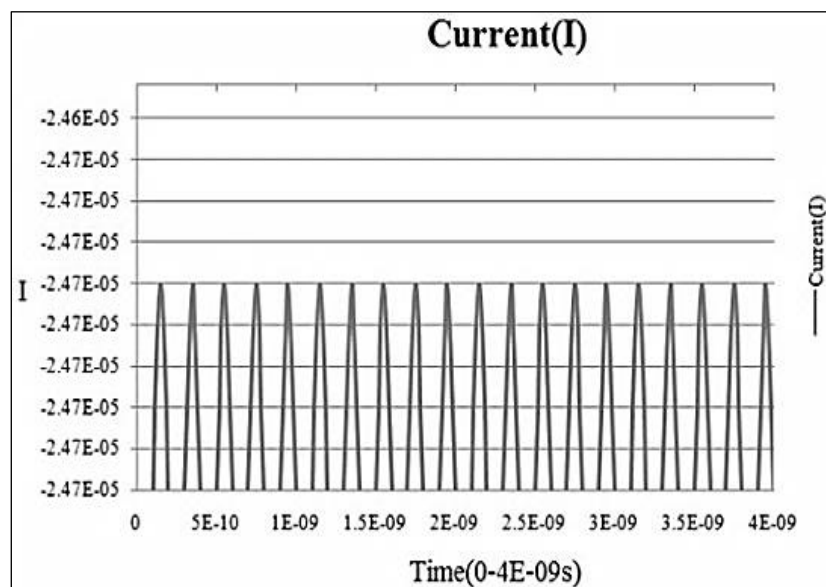


Fig. 5: Current vs. Time Characteristics where Analysis Considered up to 40 nsec [4].

Plotting voltage and the current together as shown in Figure 6 assumes analysis for 40 nsec.

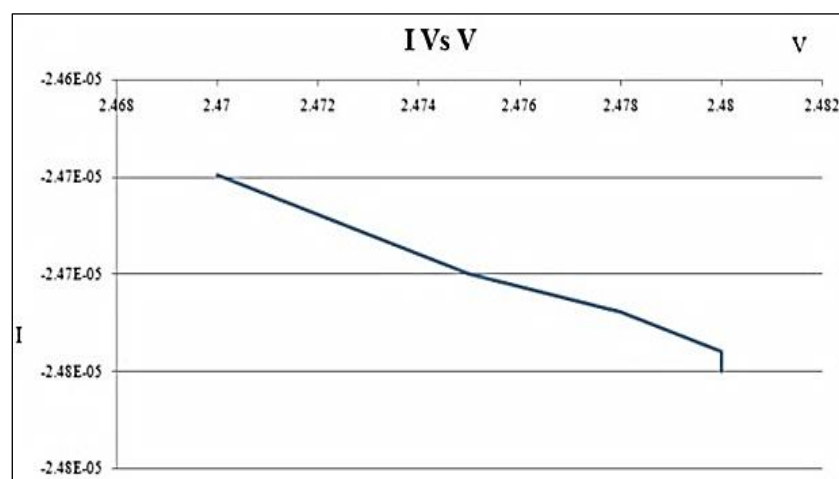


Fig. 6: Current vs. Voltages Characteristics [3].

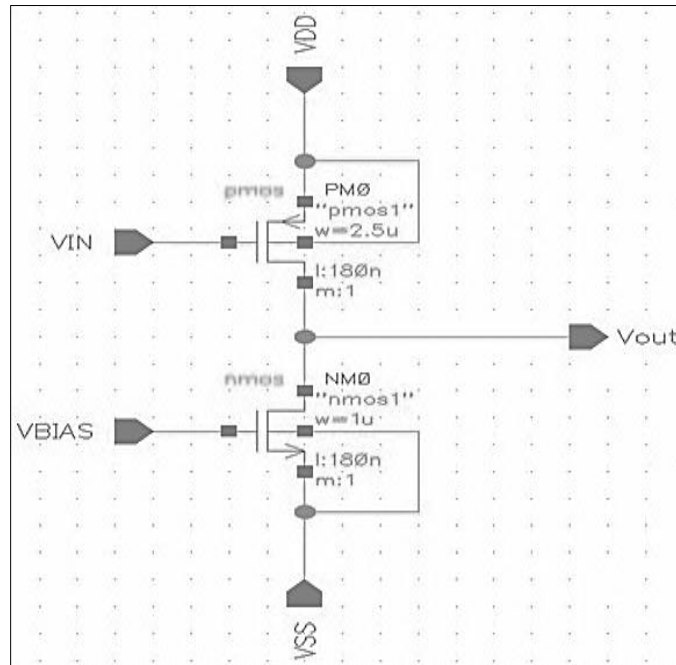
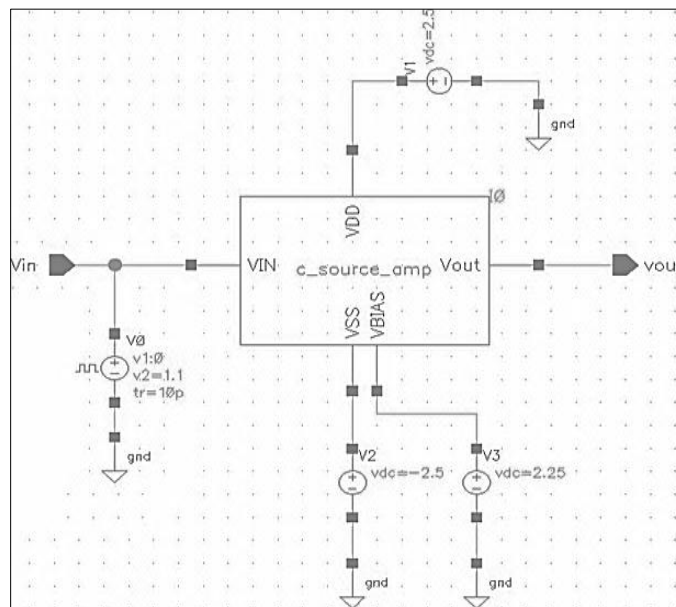
Table 1: Parameter in Transient Analysis of Memristor 0 to 40 nsec.

Parameters	Value	Parameters	Value
Starts	0 s	abstol(v)	1 uv
outputs start	0 s	abstol(i)	1 pa
Stop	40 nsec	Temp	27
Step	40 psec	Tnom	27
max step	800 psec	Tempeffects	All
Ic	All	Errpreset	Moderate
Useprevic	No	Method	Traponly
Skipdc	No	Iteration	Sigglobal
Reltol	1e-03	Cmin	0 F
Abstol (m)	10 pm	Gmin	1 ps

Parameter values shown in Table 1 defined the features of memristor during transient analysis.

COMMON SOURCE AMPLIFIER

Designed the schematic common source amplifier in the cadence virtuoso with PMOS and NMOS having width to be considered 2.5 μm and the 1 μm respectively in 180 nm channel length of MOSFET [6, 7]. Schematic presentation of a common source amplifier is shown in Figure 7.

**Fig. 7:** Schematic Presentation Common Source Amplifiers [2].**Fig. 8:** Test Bench for Common Source Amplifier [3, 4].

Perform the analysis of transient, DC and AC of common source amplifier by designed test bench as shown in Figure 8. Table 2 defines the parameters used in test bench of common source amplifier; therefore response shown in Figures 9 and 10, where invert response found in transient analysis, no affected AC analysis and decaying response found in DC analysis, we are consider Vdd, Vss and Vbias as 2.5, -2.5 and 2.25 V, respectively [8].

Table 2: Parameters in Transient, AC and DC of Common Source Amplifier.

Parameters	Value	Parameters	Value
Starts	0 sec	Temp	27
outputs start	0 sec	Tnom	27
Stop	40 nsec	Tempeffects	All
Step	40 psec	Errpreset	Moderate
max step	800 psec	Method	Trap only
Reltol	1e-03	Iteration	3.5
abstol (metr)	10 pm	Cmin	0 F
Abstol (v)	1 uv	Gmin	1 psec
Abstol (i)	1 pa		

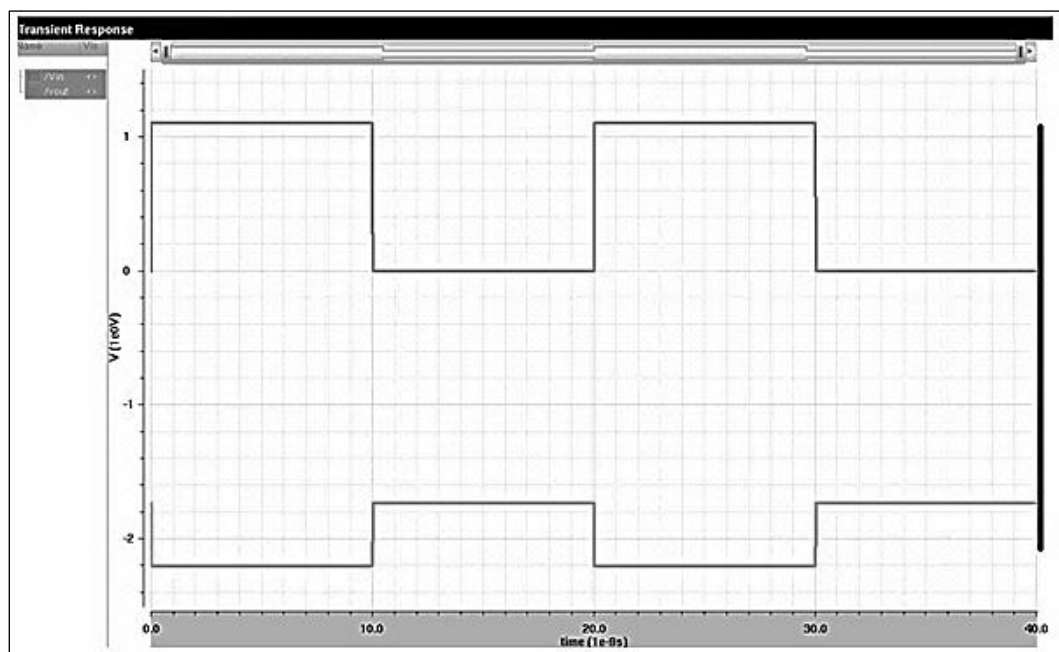


Fig. 9: Transient Response of Common Source Amplifier [2].

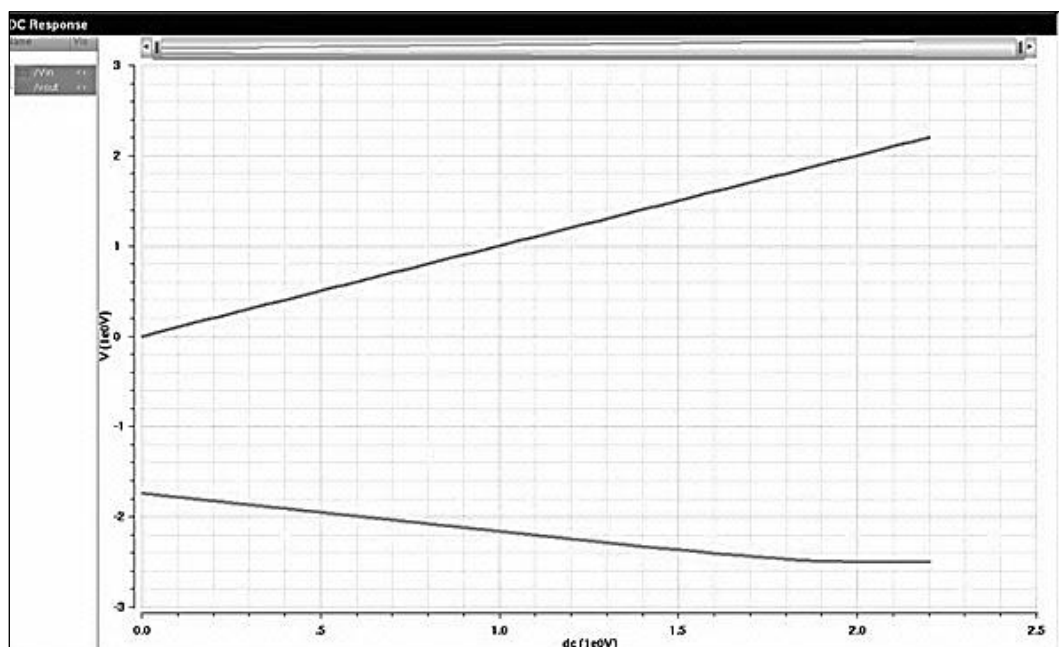


Fig. 10: DC Response of Common Source Amplifier [3].

COMMON SOURCE AMPLIFIER WITH SINGLE MEMRISTOR

Common source amplifier with single memristor at V_{in} terminal provides voltage to memristor 2.5 V with consider resistance at input terminal of memristor is $10\ \Omega$. Schematic presentation of common source

amplifier with single memristor is shown in Figure 12. Considering V_{bias} at 2.25 V and the V_{dd} , V_{ss} is 2.5, -2.5 respectively [9, 10]. The noise response of common source amplifier is shown in Figure 11 at 5 GHz frequency.

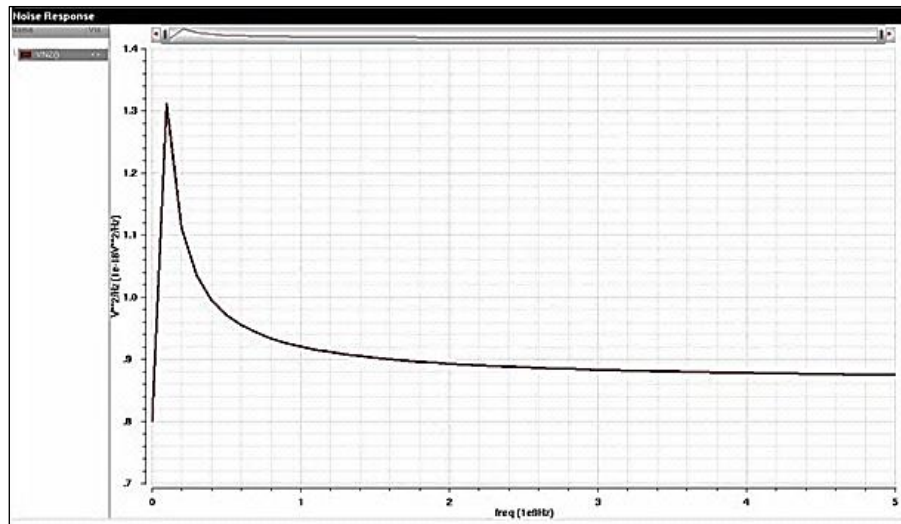


Fig. 11: Noise Response of Common Source Amplifier [3].

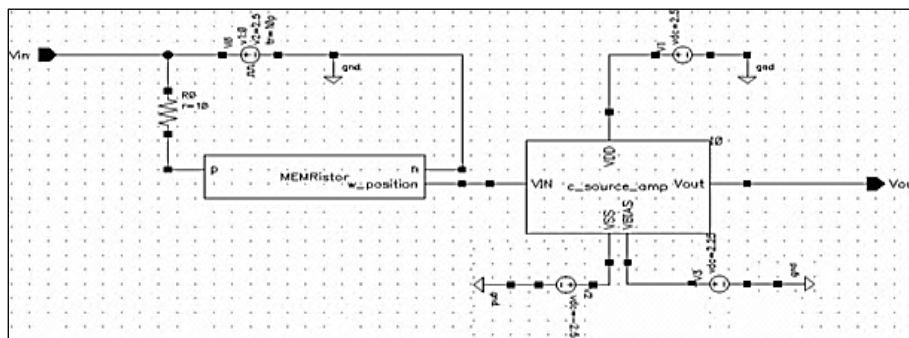


Fig. 12: Schematic Presentation of Common Source Amplifier with Single Memristor at Terminal V_{in} [2, 3].

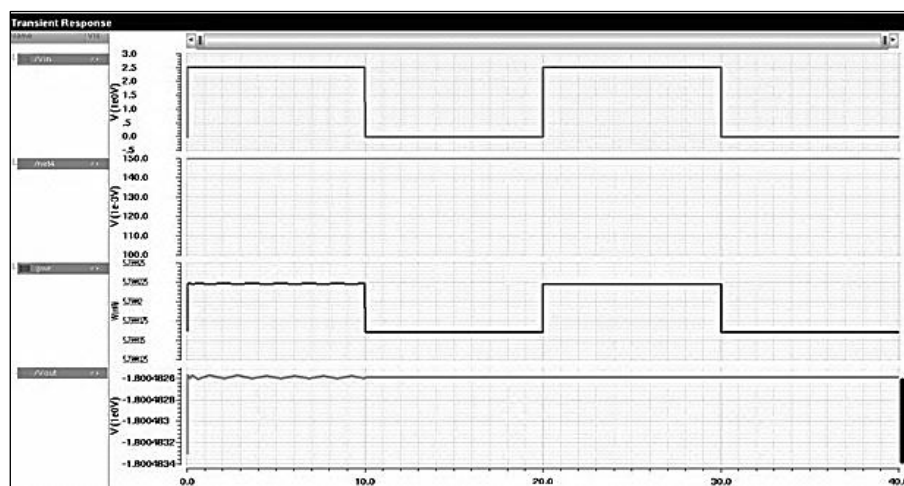


Fig. 13: Transient Analysis of Common Source Amplifier with Single Memristor [4].

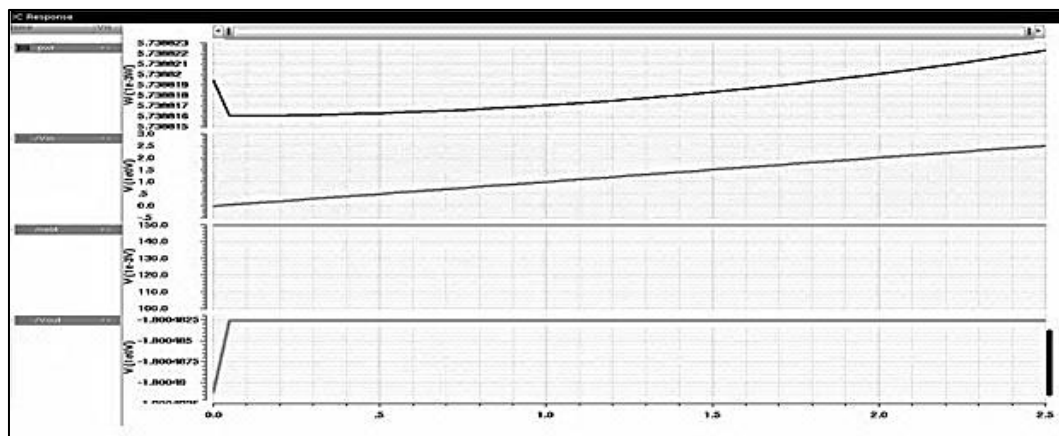


Fig. 14: DC Response of Common Source Amplifier [4].

Transient analysis of single memristor at terminal of common source amplifier as in which Voltage V_{in} , memristor output at $w_position$, power at V_o and the V_o plotting shown in Figure 13. Consider rise time and fall time as 10 psec and the pulse width is 10 nsec in period of 20 nsec and voltage is 2.5 V at V_{in} [11].

In transient analysis of common source amplifier results as average value of power is $5.739e-3$ W, maximum and minimum value of voltage is $150.0e-3$ V. In DC response analysis presents results as average value of power is $5.739e-3$ W and the analysis shown in Figure 14 [12]. Then noise response of single memristor at output terminal of common source amplifier its schematic and the response is shown in Figures 14 and 15 respectively at 5 GHz frequency, there slightly

reduced from without memristor at common source amplifier.

COMMON SOURCE AMPLIFIER WITH DOUBLE MEMRISTOR

In common source amplifier with double memristor is tends to one memristor at V_{in} terminal and the other one is at V_{bias} terminal of common source amplifier, and assume similar values like common source amplifier with single memristor approach then the authors found following characteristic analysis shown given figures. At Figure 16, shown schematic presentation of the common source amplifier with double memristor and V_{bias} terminal has also $10\ \Omega$ resistor with applied voltage 2.5 V [13].

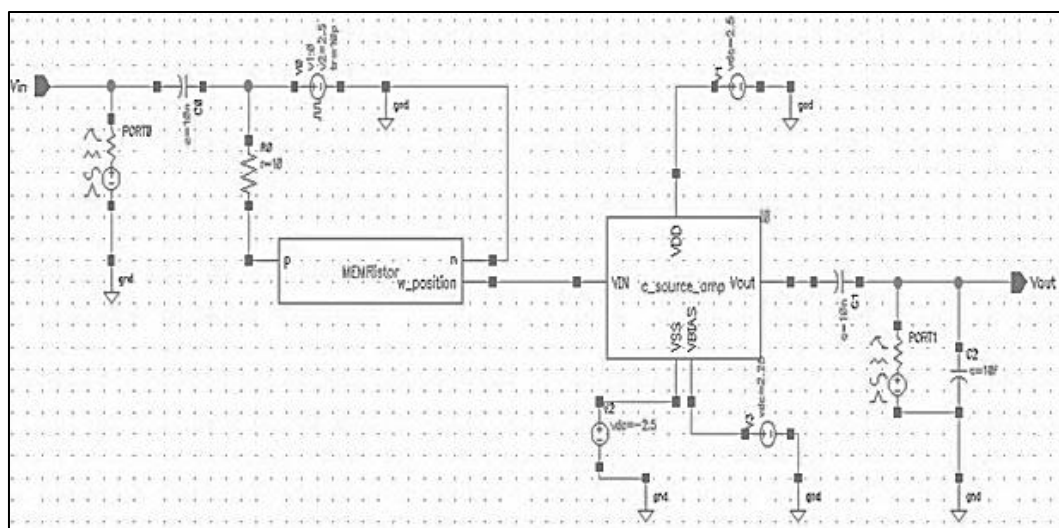


Fig. 15: Schematic Presentation of Analysis of Noise Response at Common Source Amplifier with Single Memristor [5].

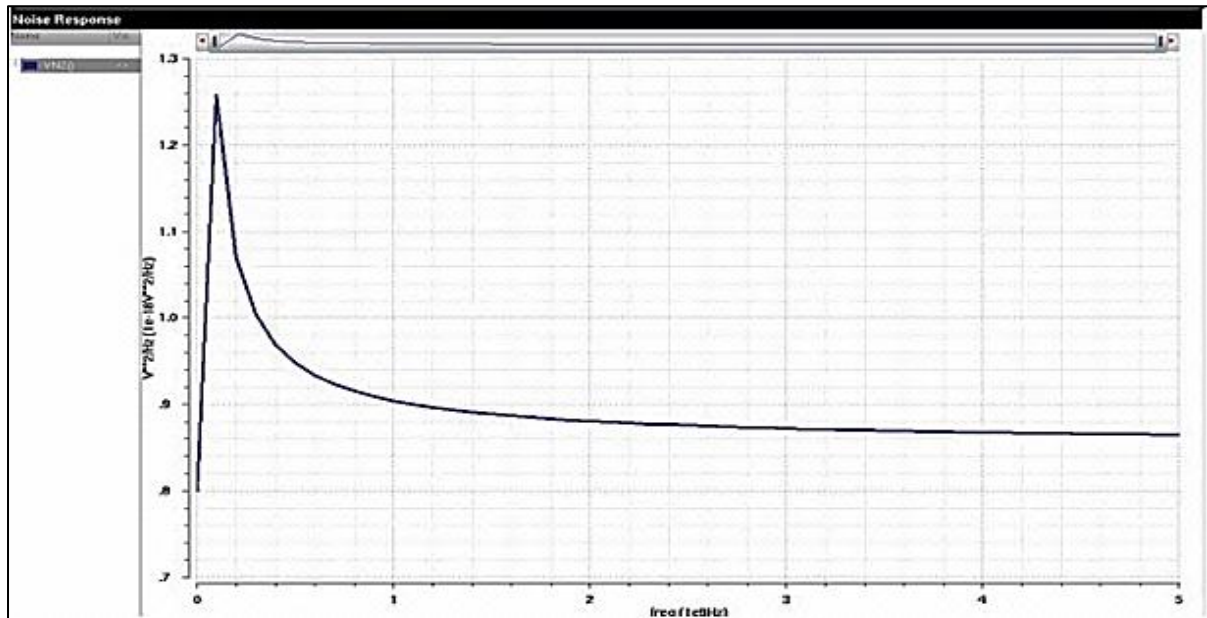


Fig. 16: Noise Response of Single Memristor at Common Source Amplifier [5].

Its DC, transient and noise response shown in Figures 17, 18 and 19, in which Vout response of transient and the DC is linear then common

source amplifier and power output is also included at Figures 20 and 21 [14, 15].

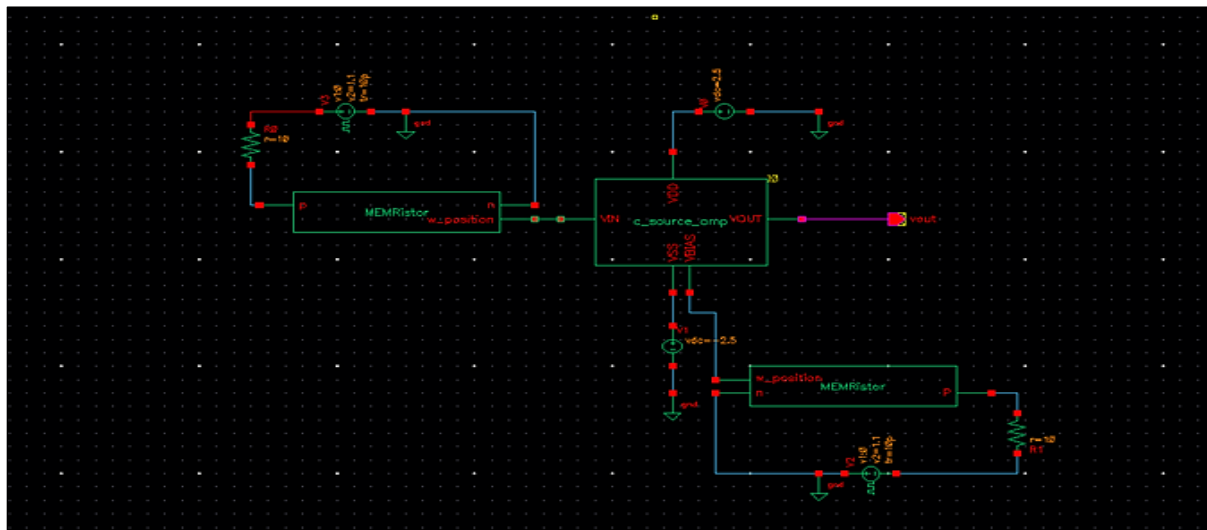


Fig. 17: Schematic Presentation of Common Source Amplifier with Double Memristor [4].

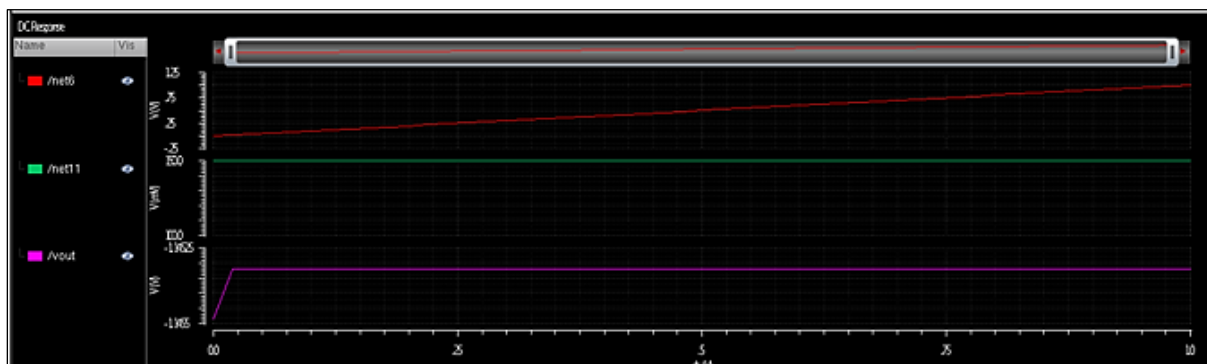


Fig. 18: DC Analysis of Common Source Amplifier with Double Memristor [4].

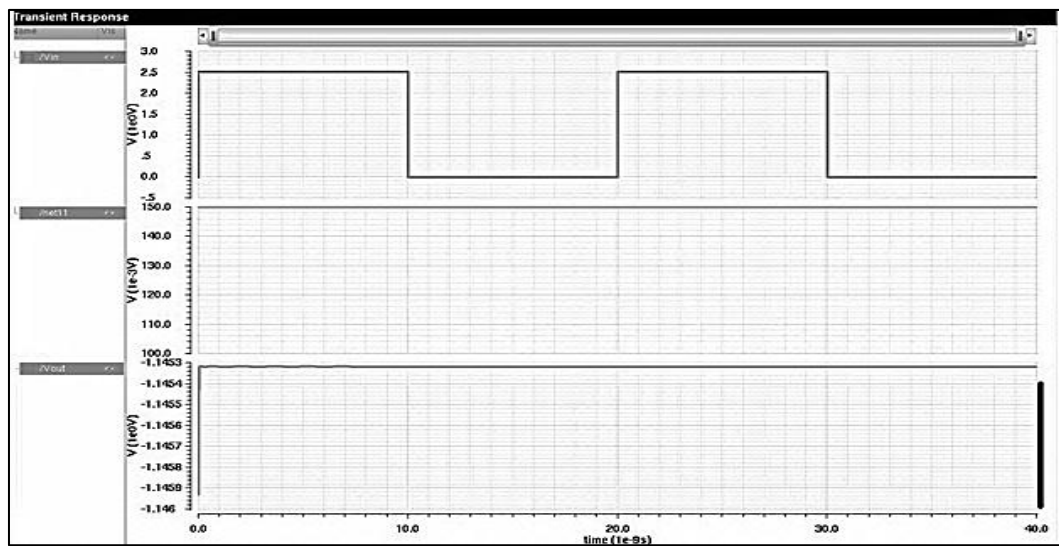


Fig. 19: Transient Analysis of Common Source Amplifier with Double Memristor [5].

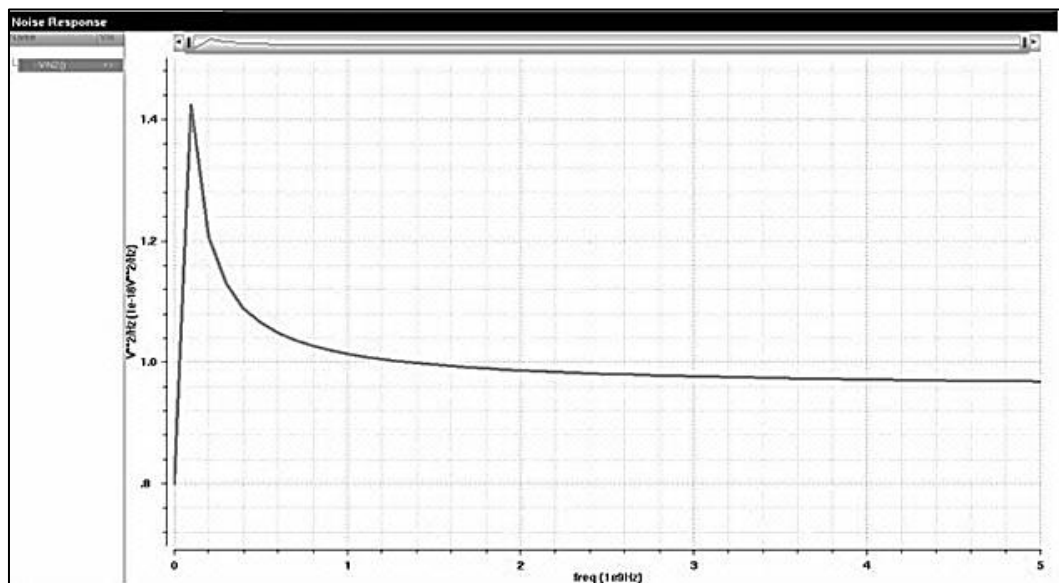


Fig. 20: Noise Response of Common Source Amplifier with Double Memristor [3, 5].

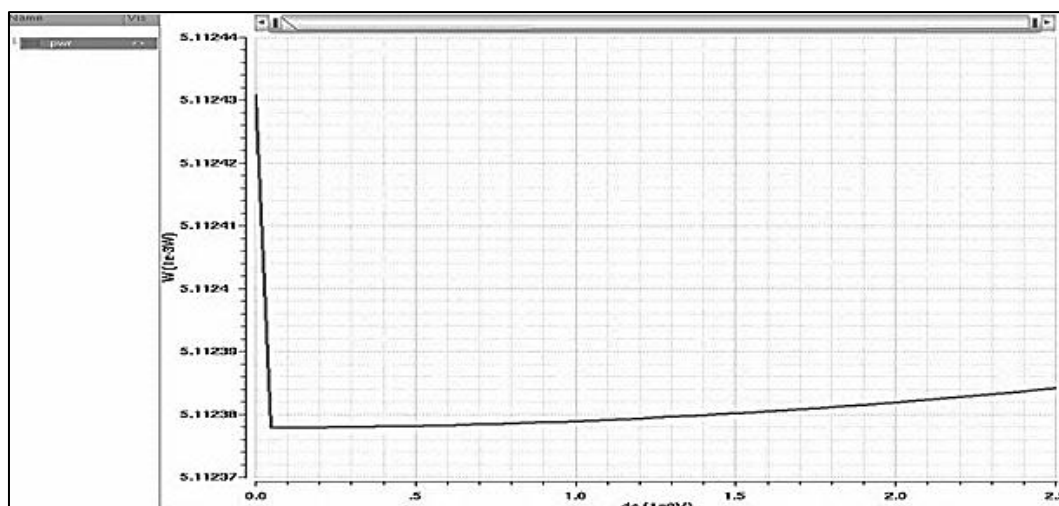


Fig. 21: Power Response of Common Source Amplifier with Double Memristor [4, 5].

CONCLUSIONS

The model of memristor in cadence virtuoso has been designed using VHDL coding; with this the values of each analysis have been represented in tabulated form to show a clear impact of the analysis. Also on performing various analyses of common source amplifier, then applying same process using single memristor common source amplifier and then again using double memristor common source amplifier in cadence virtuoso using gpdtk (generic process design kit) at 180 nm technology and finally comparing between the three processes, it has been found that with the use of memristor there is a drastic reduction in the values of noise (dB) and power dissipation (Watt) and a significant improvement in the values of transient response (comparison between voltage and time), the DC response (comparison between input and output voltages) by reducing ripple effect from response with the use of memristor. It has been proved that with the use of memristor and going on higher addition of memristor in the circuit, the circuit works efficiently with a drastic reduction in the values of noise and power respectively.

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